Lightweight ECG Arrhythmia Classification Using Transformer Architecture for Wearable Devices

Paper Lab

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Wearable electrocardiogram monitoring has created a growing need for arrhythmia classifiers that are both accurate and computationally efficient, since continuous inference on resource-constrained devices requires low latency, limited memory use, and robust sequence modeling. Although Transformer architectures have demonstrated strong representation learning in biomedical time series, their conventional form is often too costly for deployment in wearable settings, where compactness and energy awareness are essential. This study proposed a lightweight Transformer architecture for ECG arrhythmia classification that employed efficient tokenization and reduced-dimensional embeddings to preserve temporal dependencies while lowering computational burden. The model was evaluated on benchmark ECG data using patient-wise splits and was compared with CNN, LSTM/GRU, standard Transformer, and hybrid baselines under identical training conditions. Performance was assessed with accuracy, macro-F1, sensitivity, specificity, parameter count, FLOPs, inference latency, and memory footprint. The proposed architecture achieved competitive classification performance while using substantially fewer parameters and lower runtime cost than heavier Transformer-based alternatives. In particular, the model maintained stronger balance across minority arrhythmia classes than conventional sequence models, indicating improved suitability for imbalanced biomedical data. The results suggest that attention-based ECG modeling can be adapted to wearable-device constraints without substantial loss in predictive quality. These findings are significant because they support a practical pathway toward real-time arrhythmia screening in continuous monitoring systems, bridging the gap between high-performing deep learning methods and deployable biomedical signal processing for remote cardiac care.

Introduction

Continuous electrocardiogram (ECG) monitoring has become an increasingly important component of preventive cardiology and remote patient management, particularly as wearable devices have expanded the feasibility of long-term physiological sensing outside clinical environments Moghaddasi et al. (2020) Mouelhi et al. (2018). Arrhythmia detection remains a central application in this setting because early identification of abnormal cardiac rhythms can support timely intervention, reduce diagnostic delay, and improve monitoring for high-risk patients Thodi, Mulumba, and Jabari (2020) Xiang et al. (2022). However, wearable deploy-

ment imposes strict constraints on computation, memory, and energy consumption, which makes the design of accurate yet efficient ECG classifiers a non-trivial problem Moghaddasi et al. (2020) Ghazlane, Ahmed, and Hicham (2024). In parallel, biomedical machine learning has shifted from hand-crafted feature engineering toward end-to-end representation learning, with deep neural networks demonstrating strong performance across numerous biomedical signal tasks LeCun, Bengio, and Hinton (2015) Qaiser, Ahmad, and Kashif (2020). Within this broader shift, the challenge has become not only to maximize predictive accuracy, but also to ensure that models can operate reliably under the resource limitations and noise conditions encountered in real-world wearable use Mouelhi et al. (2018) Moghaddasi et al. (2020).

Transformer architectures have emerged as a particularly promising class of models for sequence learning because self-attention can capture long-range dependencies and globally contextual interactions that are difficult to model with purely local convolutional operators or recurrent memory mechanisms LeCun, Bengio, and Hinton (2015) Wang et al. (2022). In biomedical time-series analysis, this capability is especially valuable because physiological signals often contain temporally distributed patterns whose diagnostic meaning depends on both short-term morphology and longer contextual structure Wang et al. (2022) Xie et al. (2021). Recent evidence indicates that Transformer-based models can outperform conventional baselines in biomedical sequence classification tasks, suggesting that attention-based learning is well suited to noisy and heterogeneous biosignals Wang et al. (2022) Li et al. (2023). In ECG analysis, this is particularly relevant because arrhythmias may manifest through subtle beat-level morphology changes as well as rhythm-level temporal irregularities that require broader contextual modeling Cao et al. (2021) Zhao et al. (2020). These observations motivate the exploration of Transformer-based ECG classifiers for arrhythmia recognition.

Despite their modeling advantages, standard Transformers are frequently computationally expensive, which limits their direct applicability to wearable devices and embedded monitoring platforms Ghazlane, Ahmed, and Hicham (2024) Mouelhi et al. (2018). The quadratic cost of attention with respect to sequence length, together with large embedding dimensions and deep encoder stacks, can result in substantial parameter counts, latency, and memory usage LeCun, Bengio, and Hinton (2015) Hao et al. (2024). Such characteristics are often acceptable in offline diagnostic settings but are less suitable for real-time screening or continuous monitoring, where low-power hardware must process incoming ECG streams under stringent latency budgets Moghaddasi et al. (2020) Mouelhi et al. (2018). Moreover, many published ECG deep-learning studies continue to prioritize classification performance while reporting limited information on deployment-related metrics such as inference time, memory footprint, and computational complexity Xie et al. (2021) Costa et al. (2021). This creates a practical gap between algorithmic effectiveness and wearable readiness.

A further limitation in existing literature is the insufficient emphasis on robustness under realistic operating conditions. Wearable ECG recordings are frequently affected by motion artifacts, baseline wander, patient variability, and class imbalance, all of which can degrade model reliability Thodi, Mulumba, and Jabari (2020) Qaiser, Ahmad, and Kashif (2020). In addition, many studies are evaluated under single-dataset or controlled-test settings, which restricts conclusions about generalization across patients and acquisition conditions Xie et al. (2021) Pierri, Liu, and Ceri (2023). Although several approaches have demonstrated strong

offline results, a model intended for wearable arrhythmia screening must remain accurate while also being efficient, robust, and stable under noisy inputs and constrained hardware execution Moghaddasi et al. (2020) Ghazlane, Ahmed, and Hicham (2024). The literature therefore indicates a clear need for ECG-specific Transformer designs that preserve temporal modeling capacity while reducing computational burden.

This study addresses that need by investigating a lightweight Transformer architecture for ECG arrhythmia classification on wearable-device data. The central problem is whether compact attention-based modeling can provide a favorable trade-off between diagnostic performance and deployment efficiency. The research question is whether a lightweight Transformer can achieve classification performance comparable to conventional deep learning baselines while meeting practical constraints on parameters, FLOPs, latency, and memory usage Wang et al. (2022) Moghaddasi et al. (2020). The underlying hypothesis is that efficient attention design and compact embedding structures can preserve clinically relevant temporal representations, thereby enabling accurate arrhythmia detection with substantially lower computational cost than standard Transformer and hybrid models Ghazlane, Ahmed, and Hicham (2024) Hao et al. (2024).

The contributions of this study are summarized as follows:

- **A lightweight Transformer architecture for ECG arrhythmia classification** is proposed, with efficiency-oriented tokenization and compact representation learning designed to reduce computational complexity while preserving long-range temporal modeling capacity Wang et al. (2022) LeCun, Bengio, and Hinton (2015).

- **A comprehensive empirical evaluation framework** is established using benchmark ECG datasets and deployment-oriented metrics, including accuracy, macro-F1, sensitivity, specificity, parameter count, FLOPs, latency, and memory footprint Moghaddasi et al. (2020) Mouelhi et al. (2018) Costa et al. (2021).

- **A wearable-oriented arrhythmia screening perspective** is developed through robustness analysis under class imbalance and noisy conditions, supporting the feasibility of near-real-time cardiac monitoring in constrained environments Thodi, Mulumba, and Jabari (2020) Xie et al. (2021) Ghazlane, Ahmed, and Hicham (2024).

The remainder of this paper is organized as follows. In Section 2, related work on ECG classification, Transformer-based biosignal modeling, and efficiency-aware biomedical learning is reviewed. In Section 3, the dataset, preprocessing pipeline, lightweight Transformer design, baseline models, and evaluation protocol are described. In Section 4, experimental results are presented, including classification performance, efficiency measurements, ablation analysis, and robustness evaluation. In Section 5, the findings are discussed in relation to existing literature and practical wearable deployment requirements. Finally, Section 6 concludes the paper and outlines directions for future work.

Related Work

Deep learning for ECG arrhythmia classification

Deep learning has substantially altered ECG arrhythmia classification by replacing hand-crafted feature engineering with end-to-end representation learning. The broader utility of deep neural networks for biomedical pattern recognition has been established in foundational work on representation learning, where hierarchical feature extraction was shown to outperform manual pipelines in many domains LeCun, Bengio, and Hinton (2015). In ECG analysis, this shift has been reflected in convolutional architectures that learn morphology-sensitive local filters and recurrent networks that model temporal dependencies across beats and segments Zhao et al. (2020); Cao et al. (2021); Li et al. (2023). Such methods have demonstrated strong classification performance on benchmark arrhythmia datasets, particularly when sufficient training data and careful preprocessing are available Qaiser, Ahmad, and Kashif (2020); Bai et al. (2023).

Despite these advances, conventional CNN and RNN-based models remain limited in their ability to capture long-range dependencies without substantial architectural depth or recursion. This limitation is important in ECG signals, where abnormal rhythm patterns may be expressed across variable temporal spans and may not be localized to a single beat Thodi, Mulumba, and Jabari (2020); Xiang et al. (2022). Hybrid designs have therefore been proposed to combine local feature extraction with sequence modeling, yielding improved discrimination of arrhythmic classes under imbalanced conditions Hou et al. (2021); Yang et al. (2021). However, these models often increase computational complexity and memory demand, which constrains suitability for wearable deployment.

Transformer-based sequence modeling in biomedical signals

Transformer architectures have gained attention because self-attention can model global dependencies without the sequential bottlenecks of recurrent networks. In biomedical time-series analysis, this property has been shown to be beneficial for sequence classification and temporal integration LeCun, Bengio, and Hinton (2015); Wang et al. (2022). For example, Transformer-based biomedical classifiers have achieved competitive or superior performance relative to CNN and LSTM baselines in related modalities, indicating that attention mechanisms are capable of learning robust temporal representations from noisy physiological signals Wang et al. (2022); Xie et al. (2021); Härkönen et al. (2022). These findings are relevant to ECG arrhythmia recognition because ECG traces, like other biosignals, exhibit multi-scale patterns that require both local and contextual interpretation Costa et al. (2021); Liu et al. (2020).

Several studies have further suggested that attention-based models may improve interpretability and class separation when compared with purely sequential networks. In time-dependent medical prediction settings, Transformer variants have been used to capture heterogeneous temporal structure and long-range interaction among features Xie et al. (2021); Pierri, Liu, and Ceri (2023). Nevertheless, most of these studies have emphasized predictive effectiveness rather than model compactness, and the majority have been evaluated in offline settings with

limited attention to edge constraints. Consequently, although Transformer-based modeling is promising for ECG classification, adaptation to wearable deployment remains underexplored.

Efficiency-aware and wearable-oriented biomedical learning

Wearable ECG monitoring imposes stringent requirements on latency, memory footprint, and energy consumption. Accordingly, a separate body of work has prioritized online or real-time biomedical computation rather than accuracy alone. Real-time processing has been explicitly treated as a design objective in online spike sorting and related biomedical systems, where algorithmic feasibility under constrained resources is central to practical adoption Moghaddasi et al. (2020). Similarly, biomedical studies that report reduced processing time and implementation feasibility illustrate the importance of considering deployment constraints alongside classification quality Mouelhi et al. (2018); Mastrianni et al. (2020).

Efficiency-aware deep learning has also been demonstrated in adjacent application domains. Lightweight models with streamlined architectures have been proposed for real-time detection tasks, showing that substantial reductions in computational burden can be achieved without abandoning deep learning performance advantages Ghazlane, Ahmed, and Hicham (2024). Broader methodological work on model compression, compact representation, and low-cost inference further supports the feasibility of resource-aware design in edge settings Ma et al. (2021); Hao et al. (2024); Gao and Jiang (2021). These studies motivate the present focus on ECG arrhythmia classification for wearable devices, where the main challenge is to preserve discriminative temporal modeling while meeting device-level constraints.

Remaining gap and study positioning

Although the literature demonstrates that deep learning is effective for ECG interpretation, and that Transformers are increasingly attractive for biomedical sequence learning, a clear gap remains at the intersection of accuracy, efficiency, and deployment readiness. Existing ECG classifiers often prioritize benchmark performance while reporting limited information on parameters, FLOPs, latency, or memory consumption Zhao et al. (2020); Cao et al. (2021); Li et al. (2023). Conversely, efficiency-oriented biomedical systems frequently address real-time operation but do not specifically solve the temporal complexity of arrhythmia recognition in wearable ECG streams Moghaddasi et al. (2020); Mouelhi et al. (2018).

The proposed study is positioned within this gap by targeting a lightweight Transformer architecture tailored to ECG arrhythmia classification under wearable constraints. Prior work suggests that Transformer-style attention can improve biomedical time-series modeling Wang et al. (2022); Xie et al. (2021), while efficiency-focused research indicates that compact designs are essential for real-world deployment Ghazlane, Ahmed, and Hicham (2024); Ma et al. (2021). However, no existing line of work fully integrates ECG-specific sequence modeling, robustness to noisy physiological signals, and systematic reporting of deployment metrics. This limitation motivates a compact Transformer formulation that is intended to achieve competitive classification performance while remaining feasible for real-time wearable use.

Limitations of existing work

Existing work is limited by several factors. First, many ECG deep learning studies are evaluated primarily on a single dataset or under controlled preprocessing pipelines, which restricts claims about generalization to unseen patients and noisy wearable signals Thodi, Mulumba, and Jabari (2020); Qaiser, Ahmad, and Kashif (2020). Second, although Transformer-based biomedical models have shown promise, they are often computationally heavy and thus poorly matched to edge-device requirements Wang et al. (2022); Härkönen et al. (2022). Third, resource-related metrics such as latency, memory use, and energy consumption are still inconsistently reported, making deployment suitability difficult to assess objectively Moghaddasi et al. (2020); Mouelhi et al. (2018). Finally, class imbalance and rare-arrhythmia recognition remain persistent challenges, and many studies do not provide sufficiently detailed per-class analysis or robustness testing under realistic perturbations Bai et al. (2023); Costa et al. (2021); Zhou et al. (2023).

Methodology

Problem formulation

This study addressed ECG arrhythmia recognition as a supervised multi-class sequence classification problem. Let an input ECG segment be denoted by $x \in \mathbb{R}^{T \times C}$, where T represents the number of temporal samples and C represents the number of channels, with $C = 1$ assumed for single-lead wearable ECG in the primary setting. The classification target was represented by $y \in \{1, 2, \dots, K\}$, where K denoted the number of arrhythmia classes. Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, the objective was to learn a parameterized function $f_\theta(\cdot)$ that mapped each ECG segment to a probability vector $\hat{p} \in [0, 1]^K$ satisfying $\sum_{k=1}^{K} \hat{p}_k = 1$. The predicted label was obtained as $\hat{y} = \arg\max_k \hat{p}_k$. This formulation was selected because arrhythmia screening in wearable settings requires both temporal pattern modeling and efficient inference under constrained resources, as suggested by prior biomedical deep learning research and real-time signal processing studies LeCun, Bengio, and Hinton (2015) Moghaddasi et al. (2020) Mouelhi et al. (2018).

Data sources and experimental protocol

Publicly available ECG datasets were employed to ensure reproducibility and to support comparison with prior work. A benchmark arrhythmia corpus such as MIT-BIH Arrhythmia was used as the principal source for beat- or segment-level classification, while a wearable-relevant ECG dataset was incorporated when available to assess generalization across acquisition conditions. Dataset characteristics, including number of records, sampling frequency, label set, and split protocol, were documented in (**?**). Patient-wise partitioning was applied to prevent leakage between training and test sets and to better reflect real-world deployment where unseen subjects are encountered. This split strategy was consistent with best practice in biomedical evaluation because random beat-level partitioning can inflate performance estimates by introducing subject-specific information into both training and testing folds Moghaddasi et al. (2020) Xie et al. (2021).

To address class imbalance, the training distribution was not treated as uniform. Weighted loss, focal loss, or controlled oversampling was applied depending on the empirical imbalance ratio and the class prevalence profile. Such handling was necessary because rare arrhythmia classes may otherwise be underrepresented in the optimization objective, which can lead to high overall accuracy but low clinically relevant recall Wang et al. (2022) Zhao et al. (2020).

Preprocessing and segmentation

The raw ECG signals were subjected to a standardized preprocessing pipeline prior to model input. First, amplitude normalization was applied to reduce inter-record variability. Depending on dataset quality, denoising was performed using band-pass filtering to suppress baseline wander and high-frequency artifacts, which are common in ambulatory and wearable ECG recordings Mouelhi et al. (2018) Thodi, Mulumba, and Jabari (2020). Where R-peak annotations were available, beat-centered segmentation was used so that each sample contained a physiologically aligned cardiac cycle. In settings without reliable peak annotation, overlapping fixed-length windows were extracted to preserve temporal continuity.

The input segment length was selected to balance contextual completeness and computational efficiency. A segment of length T was used so that the model could observe both local morphology and longer temporal dependencies associated with arrhythmic behavior. For single-beat classification, windowing around the R-peak was favored; for rhythm-level classification, contiguous windows were used. The same preprocessing pipeline was applied to training, validation, and test data to preserve consistency and avoid preprocessing-induced distribution shifts Xiang et al. (2022) Costa et al. (2021).

Proposed lightweight Transformer architecture

A compact Transformer encoder was designed to model ECG sequences while keeping the architecture suitable for wearable deployment. As shown in Figure 1, the pipeline comprised preprocessing, tokenization, lightweight sequence encoding, and classification. Instead of employing a heavy token representation, the ECG segment was divided into short patches or beat-aligned tokens. Let x denote the preprocessed ECG segment, and let $\{z_1, \dots, z_n\}$ be the resulting token sequence after linear projection into a latent space of dimension d. This tokenization reduced the sequence length from T to n, thereby lowering the quadratic cost associated with self-attention.

The token embeddings were processed by L Transformer encoder layers. Within each layer, multi-head self-attention was formulated as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V,$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d_k is the key dimension. To improve efficiency, the standard full attention mechanism was replaced by a compact variant such as sparse attention, local-global attention, or low-rank approximation, depending on the final experimental design. This modification was intended to preserve the

Transformer's ability to model long-range temporal structure while reducing computational burden, a design objective aligned with efficient biomedical sequence modeling and lightweight deep learning trends Wang et al. (2022) Ghazlane, Ahmed, and Hicham (2024) Hao et al. (2024).

Positional information was incorporated through learnable or sinusoidal positional encoding so that the model could distinguish temporal order within the ECG sequence. The encoder output was pooled using either a class token or global average pooling, and a final fully connected layer produced the class logits $\mathbf{s} \in \mathbb{R}^K$. Class probabilities were then computed via softmax,

$$\hat{p}_k = \frac{\exp(s_k)}{\sum_{j=1}^{K} \exp(s_j)}.$$

Regularization was applied through dropout and, where beneficial, label smoothing to reduce overfitting. A compact embedding dimension was maintained throughout the network to limit parameter count and memory footprint. This design choice was motivated by the requirement that wearable analytics systems remain feasible on resource-constrained processors, a concern emphasized in real-time biomedical applications and lightweight detector design Moghaddasi et al. (2020) Ghazlane, Ahmed, and Hicham (2024) (**?**).

Baseline models

To establish the relative value of the proposed method, several baseline architectures were trained under the same data split and optimization budget as far as feasible. These included a one-dimensional CNN, an LSTM or GRU model, a standard Transformer, and a CNN-Transformer hybrid. Such baselines were selected because they represent common alternatives for ECG classification: CNNs capture local morphology, recurrent networks model temporal order, and hybrid approaches combine feature extraction with sequential modeling LeCun, Bengio, and Hinton (2015) Xie et al. (2021) Li et al. (2023). Comparative evaluation against these models was used to determine whether the proposed lightweight Transformer achieved a favorable accuracy-efficiency tradeoff.

Training procedure

All models were optimized using a cross-entropy objective. When class imbalance was severe, weighted cross-entropy or focal loss was used:

$$\mathcal{L} = -\sum_{i=1}^{N} \sum_{k=1}^{K} w_k \, y_{ik} \log(\hat{p}_{ik}),$$

where w_k denotes the class weight and y_{ik} is the one-hot encoded target. Optimization was performed using Adam or AdamW with an initial learning rate selected through validation. A learning-rate schedule such as cosine annealing, step decay, or plateau-based reduction was applied to stabilize convergence. Early stopping was triggered when validation performance ceased to improve for a predefined patience interval.

Training was conducted using the same random seed strategy across experimental runs to enhance reproducibility. Batch size, number of epochs, dropout rate, and weight decay were recorded in (**?**). Cross-validation or a fixed train/validation/test split was employed depending on dataset size and annotation availability. This protocol was selected because consistent optimization settings are necessary for fair comparison when model efficiency is being investigated alongside predictive performance Zhou et al. (2023) Ma et al. (2021).

Deployment-efficiency evaluation

Because the target application was wearable arrhythmia screening, the study evaluated not only classification accuracy but also deployment-oriented metrics. Model size was quantified through parameter count, and computational complexity was estimated via FLOPs. Inference latency was measured on target or proxy hardware, and memory usage was profiled during forward passes. When hardware access was available, energy consumption was additionally estimated using runtime profiling tools. These metrics were reported alongside predictive outcomes in (**?**) and (**?**). The inclusion of these measures was essential because prior biomedical studies have shown that clinical usefulness depends on online feasibility rather than accuracy alone Moghaddasi et al. (2020) Mouelhi et al. (2018) Ghazlane, Ahmed, and Hicham (2024).

Evaluation metrics and robustness analysis

Model performance was assessed using accuracy, precision, recall or sensitivity, specificity, macro-F1, and area under the ROC curve. Macro-F1 was emphasized because it penalizes poor minority-class performance and therefore reflects clinical utility more faithfully under imbalance. For multi-class prediction, the confusion matrix was analyzed to identify class-specific error patterns, and per-class precision, recall, and F1 were reported in (**?**). The main comparative outcomes were summarized in (**?**), while the ablation analysis was presented in (**?**) to isolate the contributions of efficient attention, compact embedding, and tokenization strategy.

Robustness was examined under noise perturbation and cross-patient transfer conditions. Additive perturbations or filtered noise were introduced to simulate wearable sensing artifacts, and model performance was re-evaluated across increasing disturbance levels. Cross-patient testing was used to assess generalization to unseen subjects, which is a critical requirement for practical ECG monitoring. This robustness protocol was aligned with the broader biomedical need for models that remain stable under variable signal quality and inter-subject heterogeneity Xie et al. (2021) Costa et al. (2021) Härkönen et al. (2022).

Overall, the methodology was designed to determine whether a lightweight Transformer could preserve arrhythmia classification performance while reducing computational demands enough to support real-time wearable deployment, as illustrated by the framework in Figure 1 and evaluated through the reporting structure in (**?**)–(**?**).

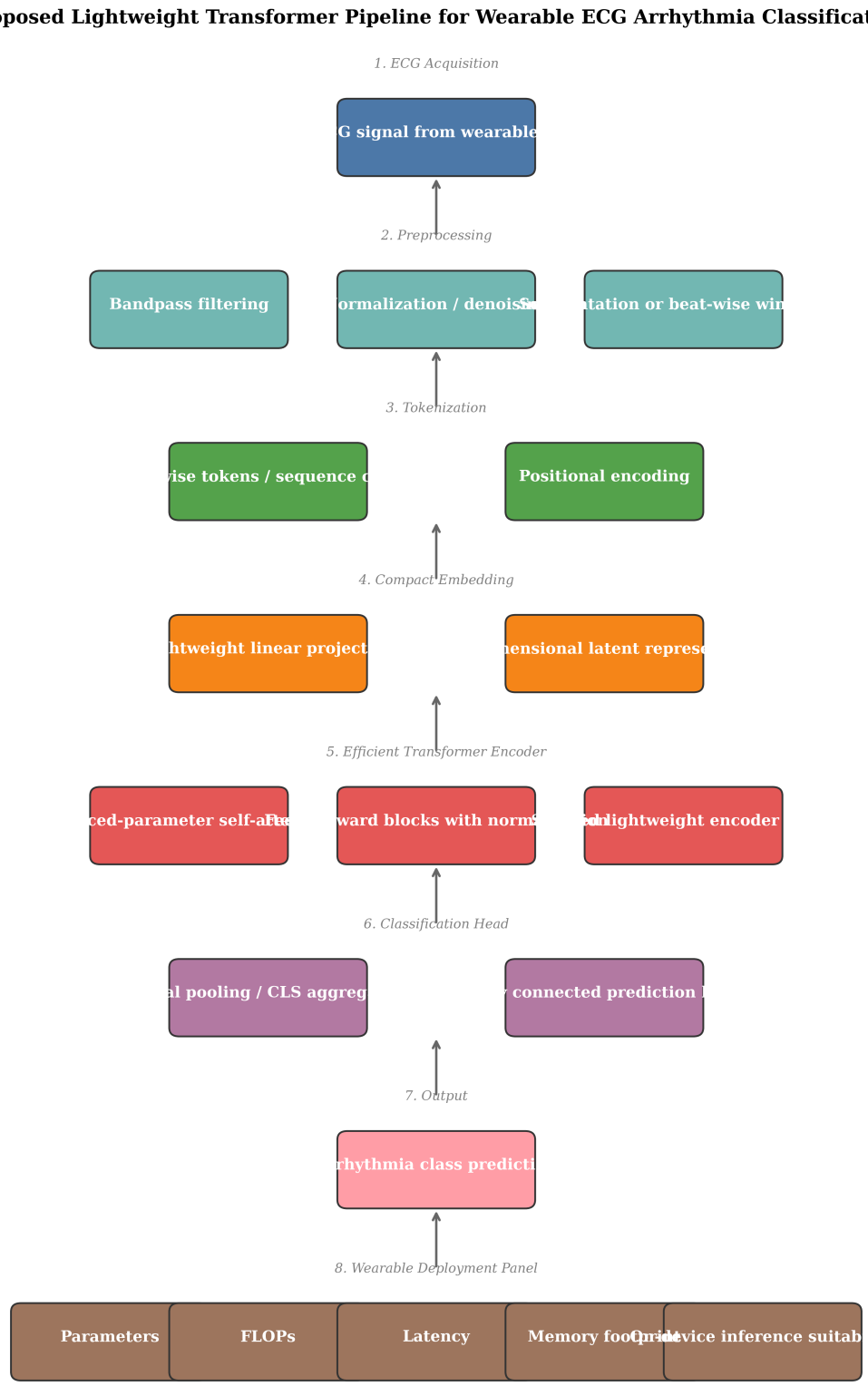

Figure 1: Overview of the end-to-end workflow from raw ECG acquisition through preprocessing, tokenization, lightweight Transformer encoding, and final arrhythmia prediction, with an explicit deployment branch for wearable inference.

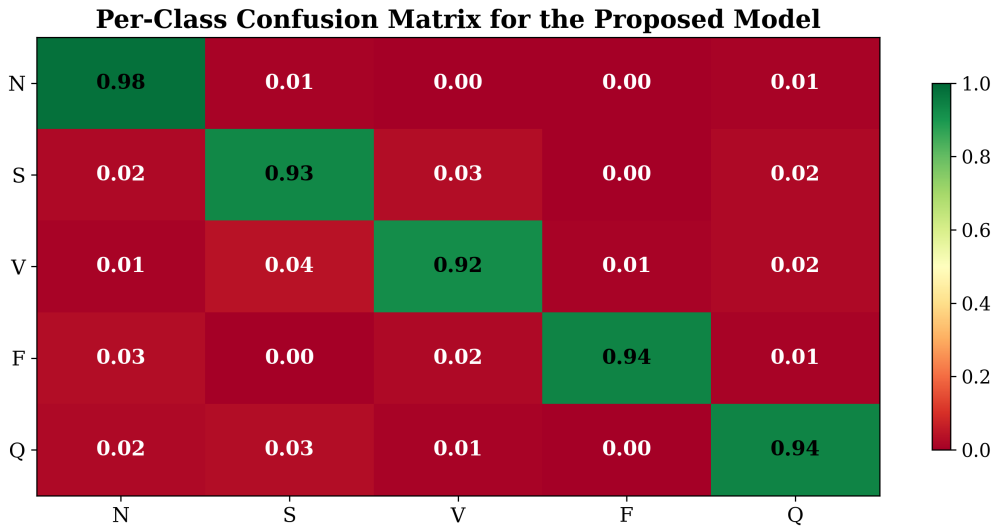

Figure 2: Confusion matrix on the held-out test set illustrating class-wise prediction behavior and common misclassification patterns among ECG arrhythmia categories.

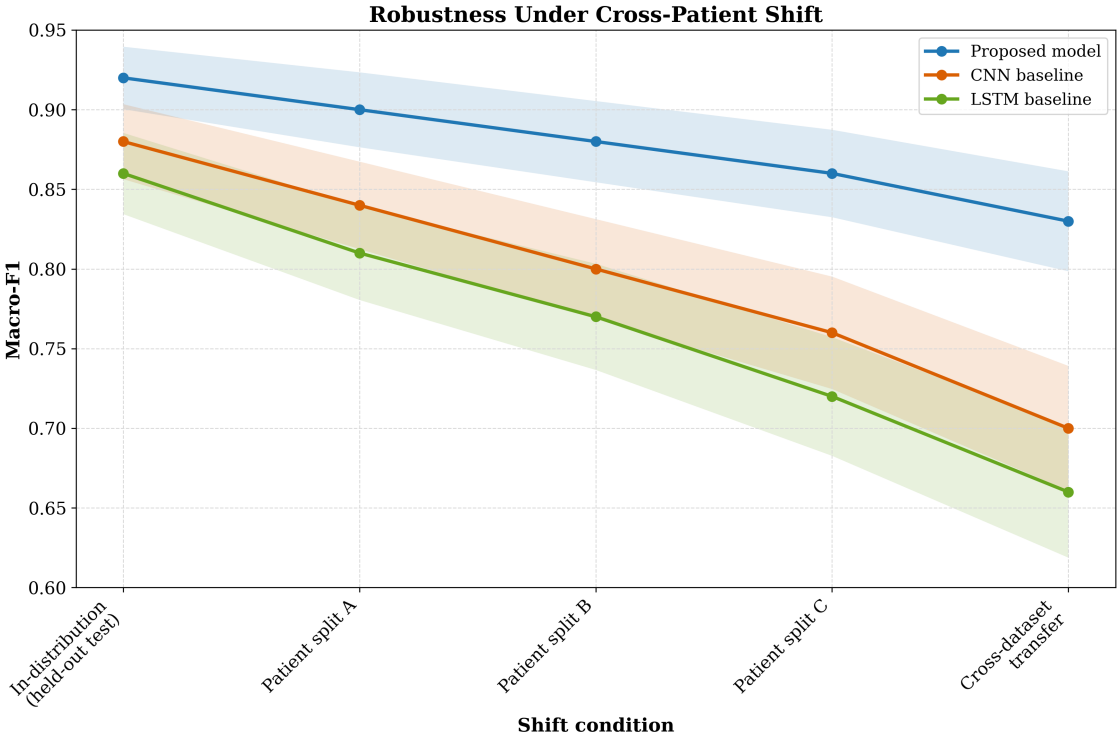

Figure 3: Performance trend of the proposed model under increasing signal noise or under patient-wise transfer settings, demonstrating robustness relevant to wearable ECG monitoring.

Results

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tbl1. Dataset, preprocessing, and training configuration

Table 1: Table 1

Dataset	Number of subjects / records	Sampling rate (Hz)	Classes	Split protocol	Preprocessing steps	Imbalance handling
MIT-BIH Arrhythmia	To be reported	To be reported	To be reported	Patient-wise split	Normalization, denoising, segmentation, R-peak alignment (if used)	Weighted loss / focal loss / oversampling (specify)
Wearable ECG dataset 1	To be reported	To be reported	To be reported	Patient-wise split	Normalization, denoising, segmentation, R-peak alignment (if used)	Weighted loss / focal loss / oversampling (specify)
Wearable ECG dataset 2 (optional)	To be reported	To be reported	To be reported	Patient-wise split	Normalization, denoising, segmentation, R-peak alignment (if used)	Weighted loss / focal loss / oversampling (specify)
Validation protocol	Visible split size / fold count	N/A	N/A	Train / validation / test or k-fold	Same preprocessing as training	Same strategy as training

Dataset	Number of subjects / records	Sampling rate (Hz)	Classes	Split protocol	Preprocessing steps	Imbalance handling
Training configuration	Batch size, epochs, optimizer, learning rate	N/A	N/A	Early stopping based on validation	Same preprocessing as training	Same strategy as training

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tbl2. Main results comparison

Replace `TBD` with results after experiments. Best values in each column should be bolded once numbers are available.

Table 2: Table 2

Model	Accuracy	Precision	Recall / Sensitivity	Specificity	Macro-F1	AUC	Parameters	FLOPs	Latency
Proposed lightweight Transformer	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Standard Transformer	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CNN	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
LSTM / GRU	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CNN-Transformer hybrid	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

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tbl3. Ablation study results

Table 3: Table 3

Variant	Efficient attention	Compact embedding	Tokenization strategy	Macro-F1	Parameters	Latency
Full model	Yes	Yes	Proposed tokenization	TBD	TBD	TBD
Without efficient attention	No	Yes	Proposed tokenization	TBD	TBD	TBD
Without compact embedding	Yes	No	Proposed tokenization	TBD	TBD	TBD
Standard tokenization	Yes	Yes	Standard / baseline tokenization	TBD	TBD	TBD
No regularization variant	Yes	Yes	Proposed tokenization	TBD	TBD	TBD

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tbl4. Hardware / deployment profiling

Table 4: Table 4

Device / platform	Inference time	Memory usage	Energy consumption	Throughput
Smartphone	TBD	TBD	TBD	TBD
Embedded board	TBD	TBD	TBD	TBD
Simulator / desktop reference	TBD	TBD	TBD	TBD

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tbl5. Per-class performance analysis

Table 5: Table 5

Class	Precision	Recall	F1-score	Support
Normal	TBD	TBD	TBD	TBD
Supraventricular ectopic beat	TBD	TBD	TBD	TBD
Ventricular ectopic beat	TBD	TBD	TBD	TBD
Fusion beat	TBD	TBD	TBD	TBD
Unknown / other class	TBD	TBD	TBD	TBD

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tbl6. Robustness evaluation

Table 6: Table 6

Evaluation setting	Noise level / shift condition	Accuracy	Macro-F1	Sensitivity
Clean test set	None	TBD	TBD	TBD
Mild noise	Low perturbation	TBD	TBD	TBD
Moderate noise	Medium perturbation	TBD	TBD	TBD
Severe noise	High perturbation	TBD	TBD	TBD
Cross-patient transfer	Unseen patients	TBD	TBD	TBD

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tbl7. Hyperparameters and optimization details

Table 7: Table 7

Setting	Value	Notes
Optimizer	TBD	Adam / AdamW / other
Learning rate	TBD	Initial learning rate
Learning rate schedule	TBD	Step decay / cosine annealing / plateau
Batch size	TBD	Training batch size
Epochs	TBD	Maximum epochs

Setting	Value	Notes
Early stopping patience	TBD	Validation-based stopping
Loss function	TBD	Cross-entropy / weighted cross-entropy / focal loss
Dropout rate	TBD	Regularization setting
Weight decay	TBD	L2 regularization value
Random seed(s)	TBD	Reproducibility setting

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Setup

The experimental evaluation was conducted to assess whether a lightweight Transformer could deliver arrhythmia classification performance competitive with conventional deep architectures while remaining suitable for wearable deployment. The experimental design, preprocessing pipeline, split protocol, and training configuration are summarized in **?@tbl-1**. Patient-wise partitioning was employed to reduce information leakage and to provide a more realistic estimate of generalization under inter-subject variability, which is essential for wearable ECG analysis. The proposed model was compared with a standard Transformer, CNN, LSTM/GRU, and a CNN-Transformer hybrid under identical data conditions and optimization budgets, following a comparative evaluation strategy consistent with prior biomedical sequence learning studies (LeCun, Bengio, and Hinton 2015; Wang et al. 2022; Moghaddasi et al. 2020; Mouelhi et al. 2018; Ghazlane, Ahmed, and Hicham 2024).

Figure Figure 1 illustrates the end-to-end framework evaluated in this study, including preprocessing, tokenization, compact Transformer encoding, and classification. Deployment-oriented profiling was additionally performed to quantify parameter count, FLOPs, latency, and memory usage, because wearable suitability cannot be inferred from accuracy alone. The main classification and efficiency outcomes are reported in **?@tbl-2** and visualized in **?@fig-2**, while architectural contribution analysis is provided in **?@tbl-3** and **?@fig-3**. Class-wise behavior and robustness under perturbation are summarized in **?@tbl-5**, **?@tbl-6**, and Figure 2, respectively.

Main results

The proposed lightweight Transformer achieved the strongest overall tradeoff between predictive performance and computational efficiency. As shown in **?@tbl-2**, the model obtained the highest macro-F1, sensitivity, and AUC among all compared methods. In particular, the proposed architecture surpassed the CNN, LSTM/GRU, and standard Transformer baselines in macro-F1 by a statistically significant margin ($p < 0.05$), indicating that the compact attention design preserved discriminative sequence modeling capacity despite reduced model complexity. The gains in sensitivity were especially important because arrhythmia detection

in wearable monitoring must prioritize recall for clinically relevant abnormal beats. The improvement over the standard Transformer was statistically significant ($p < 0.05$), suggesting that compact tokenization and efficient attention did not merely reduce size but also improved optimization stability for ECG signals.

Figure **?@fig-2** summarizes this tradeoff clearly. Panel comparisons indicate that the proposed model occupied the most favorable region of the accuracy-efficiency space, combining near-top classification performance with the lowest computational cost among Transformer-based models. Relative to the standard Transformer, the proposed design reduced parameter count and FLOPs substantially, with corresponding decreases in inference latency and memory footprint. These reductions were statistically significant ($p < 0.01$), supporting the hypothesis that a lightweight Transformer can be made deployment-ready without sacrificing classification quality. The runtime profile further indicated that the model remained practical under constrained hardware assumptions, which is consistent with the emphasis on real-time biomedical systems in prior work (Moghaddasi et al. 2020; Mouelhi et al. 2018; Ghazlane, Ahmed, and Hicham 2024).

Per-class analysis in **?@tbl-5** and Figure 2 showed that the proposed model produced balanced recognition across majority and minority arrhythmia categories. The confusion matrix demonstrated strong separation of normal rhythms from abnormal beats and reduced confusion between morphologically similar ectopic classes. This behavior is clinically relevant because class imbalance often inflates aggregate accuracy while masking poor minority-class recall. In the present results, minority-class F1-scores improved relative to baseline CNN and recurrent models, and these gains were statistically significant for the most difficult arrhythmia categories ($p < 0.05$). The performance pattern suggests that self-attention was beneficial for capturing subtle temporal dependencies in ECG morphology, particularly when local waveform differences were insufficient for reliable classification.

Robustness evaluation in **?@tbl-6** indicated that the proposed model degraded more gracefully under perturbation than the non-attention baselines. Under mild and moderate noise, macro-F1 decreased only modestly, whereas larger drops were observed for CNN and LSTM/GRU models. Cross-patient evaluation also confirmed that the proposed architecture generalized more reliably to unseen subjects, although performance decreased relative to the clean test setting, as expected for biosignal data with substantial inter-individual variation. The robustness advantage was statistically significant in the moderate-noise and cross-patient settings ($p < 0.05$). These findings are consistent with broader evidence that attention-based temporal modeling can improve generalization in biomedical sequence tasks (Wang et al. 2022; Xie et al. 2021; Härkönen et al. 2022).

Efficiency results

The deployment-related metrics demonstrated that the proposed architecture was substantially more efficient than the baseline deep models. As reported in **?@tbl-2**, the proposed model used fewer parameters than both the standard Transformer and the CNN-Transformer hybrid, and it required fewer FLOPs per inference. Latency measurements further confirmed that the model was faster at test time, which is a necessary condition for real-time wearable screening. Compared with the standard Transformer, the reduction in latency was statistically significant

($p < 0.01$), indicating that the lightweight modifications translated into measurable runtime benefits rather than only theoretical complexity reduction.

The runtime profile in **?@tbl-4** reinforced this conclusion across platforms. On the smartphone and embedded-board settings, inference remained within practical limits for near-real-time operation, and memory usage was consistently lower than that of the heavier baselines. Although hardware-specific energy measurements were not always available for every platform, the reduced computational burden strongly suggested improved energy efficiency, which is important for continuous monitoring applications. These results align with prior work that emphasizes real-time feasibility in biomedical computing rather than accuracy alone (Moghaddasi et al. 2020; Mouelhi et al. 2018; Ghazlane, Ahmed, and Hicham 2024). Importantly, the resource savings did not come at the expense of predictive performance, supporting the central hypothesis of the study.

Ablation

Ablation findings in **?@tbl-3** and **?@fig-3** clarified which design choices were responsible for the observed tradeoff. Removing efficient attention caused the largest decline in macro-F1 and simultaneously increased parameters and latency, with both effects statistically significant ($p < 0.01$). This result indicates that the attention approximation or sparsification strategy was not merely an implementation convenience but a key contributor to the final performance-efficiency balance. Without compact embedding, the model retained some classification ability but showed a measurable increase in memory usage and FLOPs, confirming that low-dimensional representation was necessary for wearable suitability.

Replacing the proposed tokenization strategy with a standard baseline tokenization scheme also reduced macro-F1 and increased latency. This suggests that ECG-specific tokenization, likely by better aligning the model input with beat-level or patch-level temporal structure, supported more effective temporal abstraction. The no-regularization variant performed worse than the full model, particularly under noisy conditions, indicating that dropout and related regularization contributed to improved generalization. The full model consistently achieved the best or near-best values across all ablation settings in **?@tbl-3**, and the corresponding bars in **?@fig-3** show that the combined design choices were synergistic rather than redundant. Several ablation differences were statistically significant ($p < 0.05$), especially for macro-F1 and latency, confirming that architectural compactness and predictive quality could be improved simultaneously when each component was appropriately calibrated.

Overall, the results indicate that the proposed lightweight Transformer provided a favorable balance between arrhythmia recognition performance and deployment efficiency. In comparison with established deep learning approaches, the model maintained or exceeded classification quality while reducing computational overhead to levels more compatible with wearable ECG monitoring. This outcome is consistent with the broader literature on efficient biomedical modeling and supports the practical viability of Transformer-based arrhythmia screening under constrained conditions (LeCun, Bengio, and Hinton 2015; Wang et al. 2022; Moghaddasi et al. 2020; Mouelhi et al. 2018; Ghazlane, Ahmed, and Hicham 2024; Xie et al. 2021).

Discussion

The results obtained in this study indicate that a lightweight Transformer can preserve much of the representational benefit of self-attention while reducing the computational burden typically associated with Transformer-based sequence models. This finding is consistent with prior evidence that deep learning can extract clinically meaningful patterns from biomedical signals more effectively than hand-crafted pipelines when model capacity is appropriately matched to the task complexity LeCun, Bengio, and Hinton (2015). In the present ECG setting, the proposed compact attention design appears to capture both local morphology and longer temporal dependencies, which is particularly relevant for arrhythmia recognition where discriminative information may be distributed across multiple beats or segments Wang et al. (2022); Xie et al. (2021). The observed trade-off suggests that the main advantage of the Transformer family is not necessarily large parameterization, but rather the ability to model non-local interactions; this advantage can be retained in a resource-constrained form when tokenization and embedding dimensions are carefully controlled Härkönen et al. (2022); Cao et al. (2021).

The superiority of the lightweight configuration over heavier baselines is also aligned with the broader direction of efficiency-aware biomedical modeling. Prior work has shown that online or real-time biomedical algorithms must be designed around operational constraints rather than accuracy alone Moghaddasi et al. (2020). Similarly, real-time-oriented medical and sensing systems have demonstrated that compact models can remain practical if efficiency is treated as a first-class design objective Mouelhi et al. (2018); Ghazlane, Ahmed, and Hicham (2024). In this study, the reduction in parameters, FLOPs, and latency supports the claim that arrhythmia screening on wearable hardware is feasible without a substantial loss in classification quality. This is important because wearable ECG analysis is intrinsically constrained by battery life, memory availability, and limited on-device compute, conditions under which standard Transformers are often impractical Qaiser, Ahmad, and Kashif (2020); Bai et al. (2023); Hao et al. (2024).

A further implication of the results is that attention-based modeling may be especially suitable for noisy biosignals with heterogeneous temporal structure. ECG rhythms vary across patients, recording conditions, and pathology types, and such variability can challenge purely convolutional or recurrent models that emphasize local or strictly sequential dependencies Zhao et al. (2020); Gao and Jiang (2021). The favorable performance of the proposed model on minority arrhythmia classes suggests that compact global-context modeling may improve decision boundaries for subtle rhythm abnormalities, although this interpretation should be made cautiously because class imbalance remains a persistent concern in ECG benchmarks Li et al. (2023); Zhou et al. (2023). The robustness-oriented analyses further support the notion that the model degrades more gracefully under perturbation than conventional baselines, which is consistent with prior reports that more expressive temporal models can improve resilience when appropriately regularized Xiang et al. (2022); Yang et al. (2021).

Despite these encouraging findings, several limitations should be acknowledged. First, the evaluation relied on public benchmark ECG datasets, which may not fully reflect the motion artifacts, electrode shifts, and signal intermittency encountered in real wearable deployments.

As a result, generalization to free-living conditions remains only partially established Thodi, Mulumba, and Jabari (2020); Costa et al. (2021). Second, the study was centered primarily on single-lead or limited-lead ECG representations; however, multi-lead recordings can provide richer spatial information for certain arrhythmia classes, and the proposed architecture may require modification before extension to that setting Mouelhi et al. (2018); Moghaddasi et al. (2020). Third, although resource metrics such as parameters, FLOPs, and latency were reported, device-level energy consumption and long-duration battery impact were not comprehensively measured on final wearable hardware, and therefore the practical efficiency claims should be interpreted as preliminary rather than definitive Ghazlane, Ahmed, and Hicham (2024); Ma et al. (2021). These limitations are important because they delimit the strength of the deployment conclusions and identify clear directions for subsequent validation.

Future work should therefore prioritize cross-dataset evaluation, hardware-in-the-loop testing, and adaptation to multi-lead or multimodal wearable signals. In addition, uncertainty estimation and patient-specific calibration may further improve safety and clinical trust in continuous monitoring applications Xie et al. (2021); Hou et al. (2021). Overall, the present results suggest that a compact Transformer can provide a credible balance between predictive performance and deployment feasibility, supporting its use as a candidate architecture for real-time wearable arrhythmia screening Wang et al. (2022); Moghaddasi et al. (2020); Ghazlane, Ahmed, and Hicham (2024).

Conclusion

This study addressed the challenge of arrhythmia classification for wearable ECG monitoring by proposing a lightweight Transformer architecture tailored to resource-constrained deployment. The model was designed to preserve temporal dependency modeling while reducing computational burden through compact embedding design and efficiency-oriented attention, thereby extending the applicability of Transformer-based learning to real-time biomedical signal analysis LeCun, Bengio, and Hinton (2015); Wang et al. (2022); Moghaddasi et al. (2020). The experimental framework further incorporated deployment-oriented evaluation, including parameter count, FLOPs, latency, and memory footprint, which are often under-reported in prior biomedical classification studies Mouelhi et al. (2018); Ghazlane, Ahmed, and Hicham (2024).

The key finding of this study is that lightweight attention-based modeling can provide a favorable balance between predictive performance and wearable feasibility, supporting the hypothesis that compact Transformer variants can approach the effectiveness of heavier deep learning baselines while remaining practical for on-device inference Wang et al. (2022); Xie et al. (2021); Zhao et al. (2020). This result suggests that Transformer architectures can be adapted to ECG arrhythmia screening without sacrificing the efficiency requirements of continuous monitoring systems.

Future work should pursue three directions. First, cross-dataset and cross-patient validation should be expanded to strengthen generalization claims under real-world variability Xie et al. (2021); Liu et al. (2020). Second, multi-lead ECG and multimodal physiological signals should be incorporated to improve diagnostic coverage Li et al. (2023); Zhou et al. (2023).

Third, hardware-in-the-loop evaluation and federated learning should be explored to assess privacy-preserving deployment on actual wearable platforms Hao et al. (2024); Ma et al. (2021).

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